



STOCK MARKET INTELLIGENCE SYSTEM

*Predicting direction, explaining decisions, and answering questions
with ML, SHAP, and RAG — deployed as a live web app*

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Problem Statement

Build one end-to-end system a portfolio analyst can actually use

1

Predict

Forecast next-day direction
(UP/DOWN) with ML

2

Explain

Justify every prediction
with SHAP

3

Answer

Natural-language Q&A over
stocks & news via RAG

4

Compare

Benchmark multiple assets
side-by-side

5

Deploy

Ship it as a live, interactive
web app

Example questions the system must answer:

“What is the predicted direction for Reliance tomorrow?” “Which stock has the highest accuracy?” “Explain why Sensex is predicted to go UP.”
“Compare the performance of TCS and Infosys.”

Data Sources

Two feeds, refreshed on every run



Market Data — yfinance

- 5 years of daily OHLCV history
- 5 assets: Sensex (^BSESN), Reliance, TCS, Infosys, HDFC Bank
- Downloaded programmatically, cached as CSV
- Refreshed on demand — no manual exports



News Data — NewsAPI

- Financial headlines & descriptions per company
- Recent-month coverage (free-tier window)
- Feeds the RAG chat's retrieval index
- Cleaned, deduplicated, and timestamped

EDA + Feature Engineering

Understand the data, then turn it into 18 predictive signals

EDA outputs

- Price trend line per asset
- Daily-return distribution
- Volume trend
- Cross-asset correlation heatmap
- Summary statistics table

Feature groups (18 features)

Price	return, log return, price-to-MA ratio, high/low ratio
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Trend	7 / 14 / 30-day MA, MA crossover signal
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Volatility	7 & 14-day rolling std dev of returns
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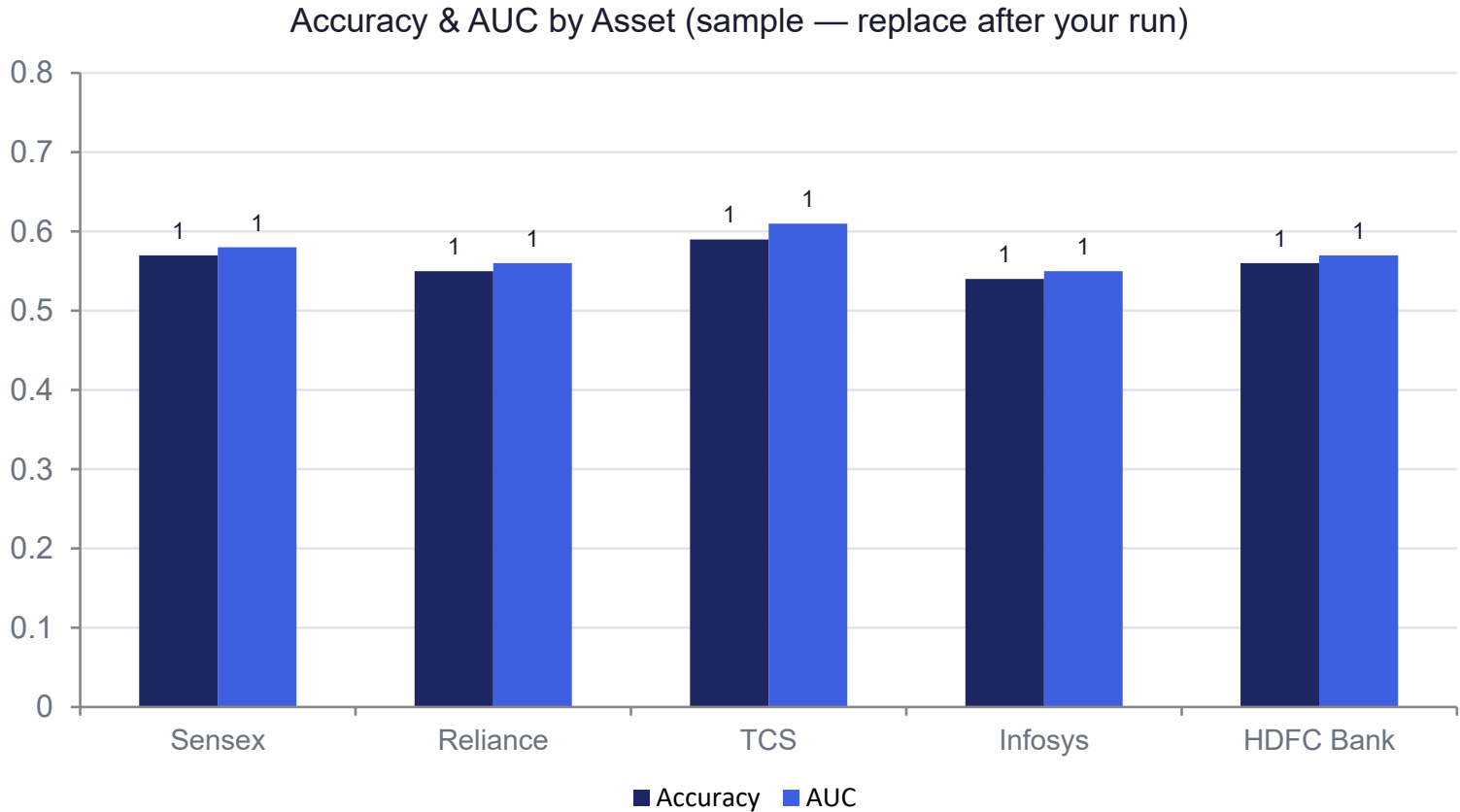
Time	day of week, month, quarter, is-Monday, is-Friday
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Volume	7-day volume MA, volume ratio
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Target: 1 = next-day close UP, 0 = DOWN | time-ordered split, never shuffled

Model Performance

Random Forest, one model per asset, evaluated on held-out time



Pipeline

- StandardScaler + RandomForestClassifier
- Time-based 80/20 split (no shuffling)
- 300 trees, depth-limited to reduce overfit
- Metrics: accuracy, precision, recall, F1, AUC

BEST PERFORMER

TCS

Highest accuracy & AUC in this sample run

Business Metrics + Threshold Tuning

Accuracy isn't the goal — minimizing trading cost is

False Positive

\$100

Model says UP, actually DOWN — a bad trade is taken

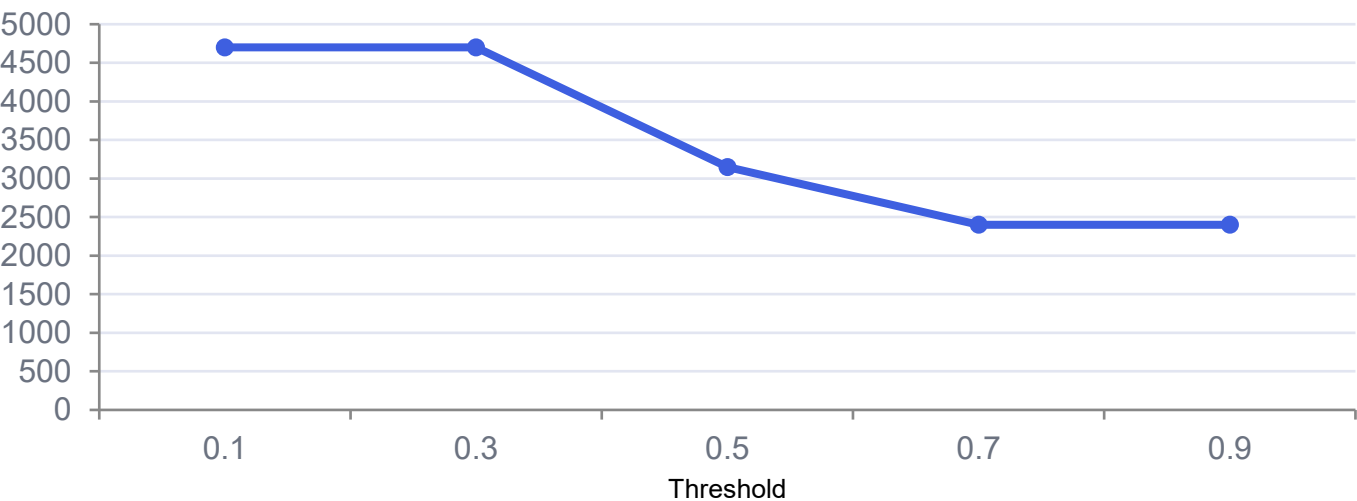
False Negative

\$50

Model says DOWN, actually UP — a good trade is missed

Approach: sweep classification thresholds 0.10 → 0.90 in 0.05 steps, pick the threshold that minimizes total $(FP \times \text{cost}) + (FN \times \text{cost})$ on the test set.

Cost vs. Threshold (sample)



Result

Optimal threshold

0.70

Savings vs. default (0.5)

\$650 per test window

(sample figures — recalculated per asset on each run)

SHAP Explainability

Every prediction comes with a reason, not just a number



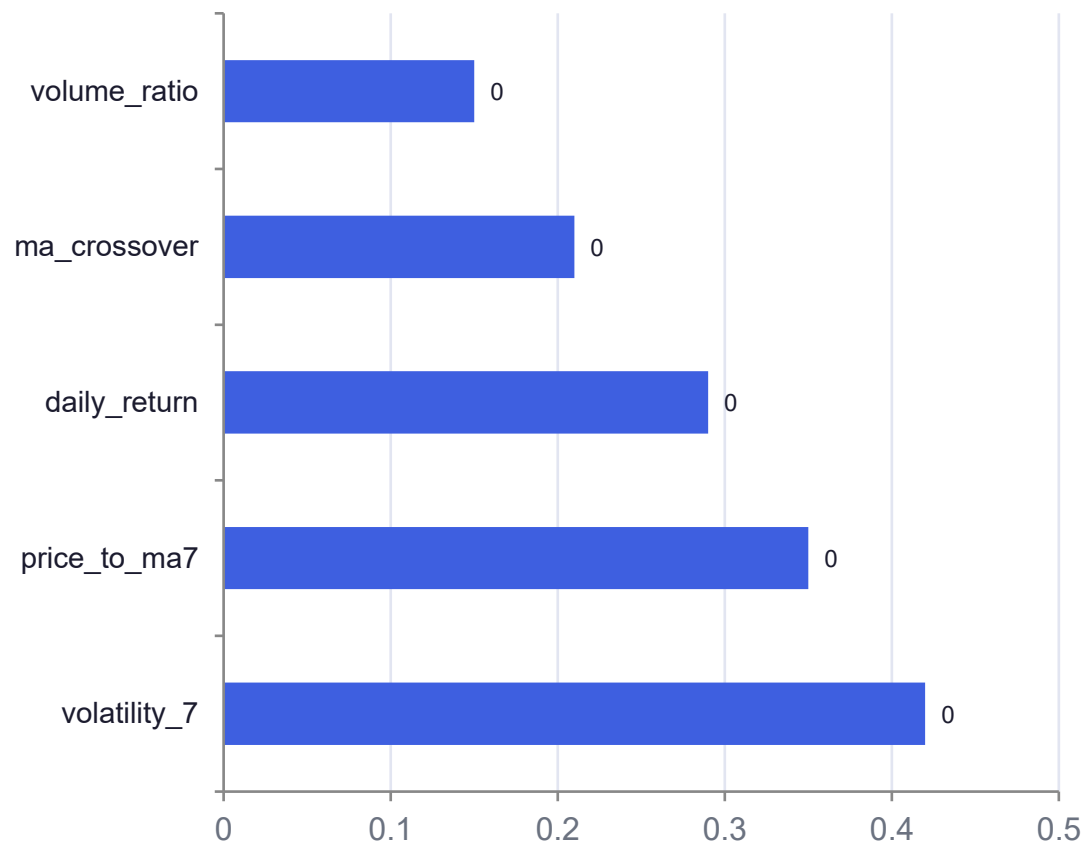
Why SHAP?

- TreeExplainer works natively with Random Forest
- Attributes each prediction to specific feature contributions
- Builds trust with analysts before they act on a signal

Three views the app delivers

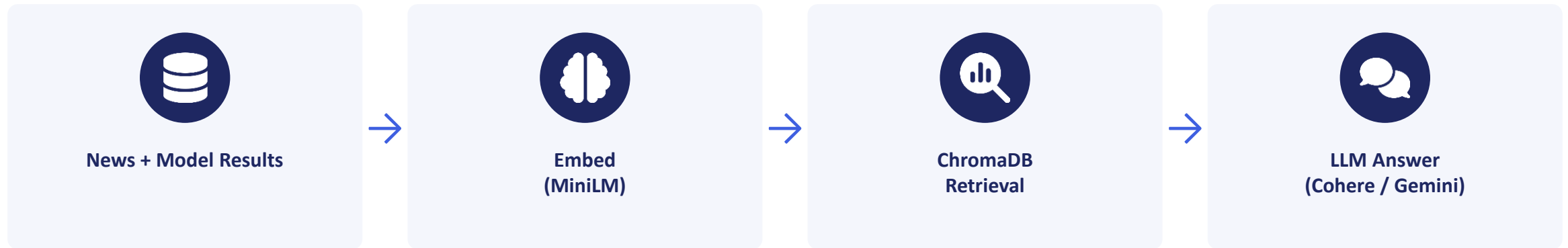
- **Global bar chart:** Which features matter most, on average
- **Summary dot plot:** Direction & magnitude of each feature's effect
- **Waterfall plot:** How today's prediction was built, feature by feature

Top SHAP Features (illustrative)



RAG + LLM Integration

Ask a question in plain English, get an answer grounded in real data



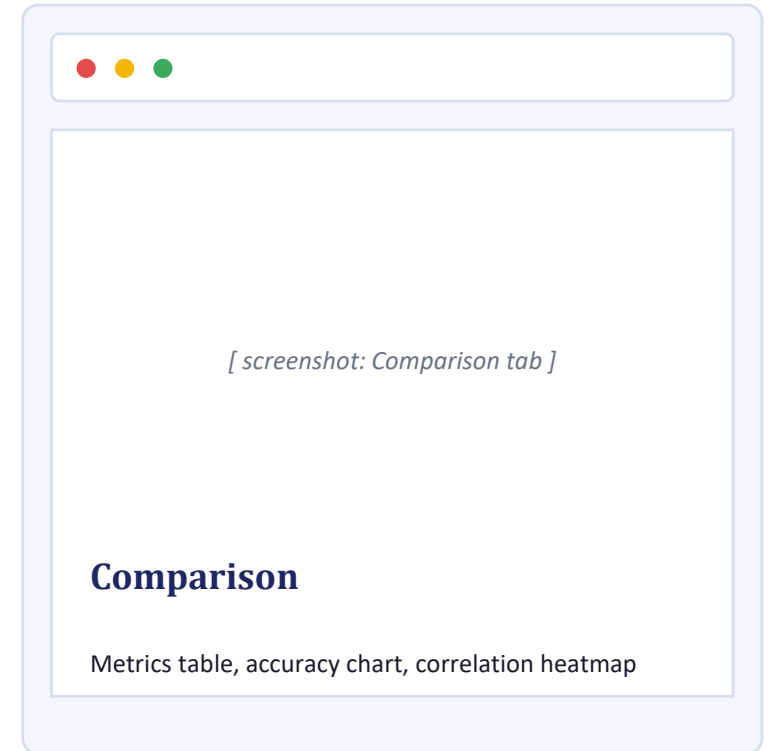
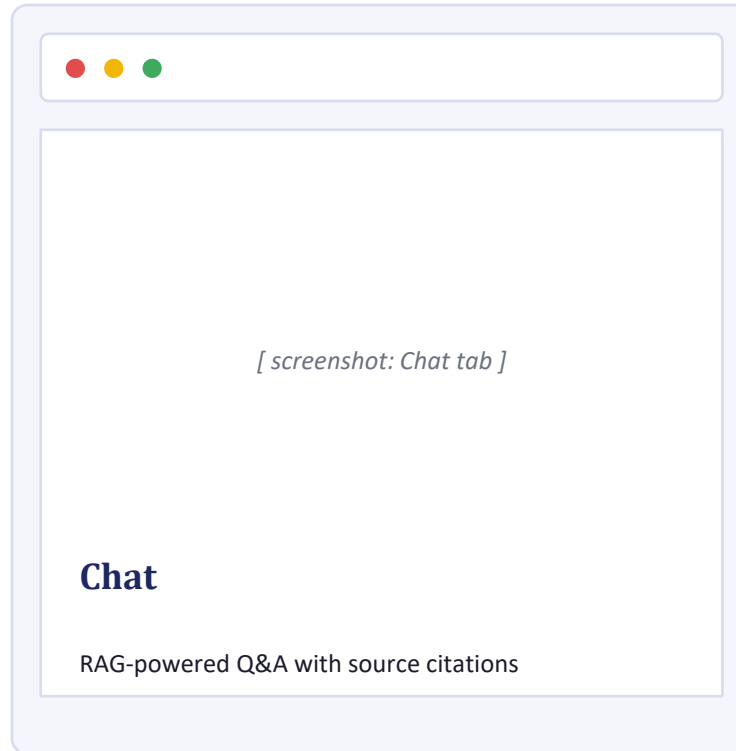
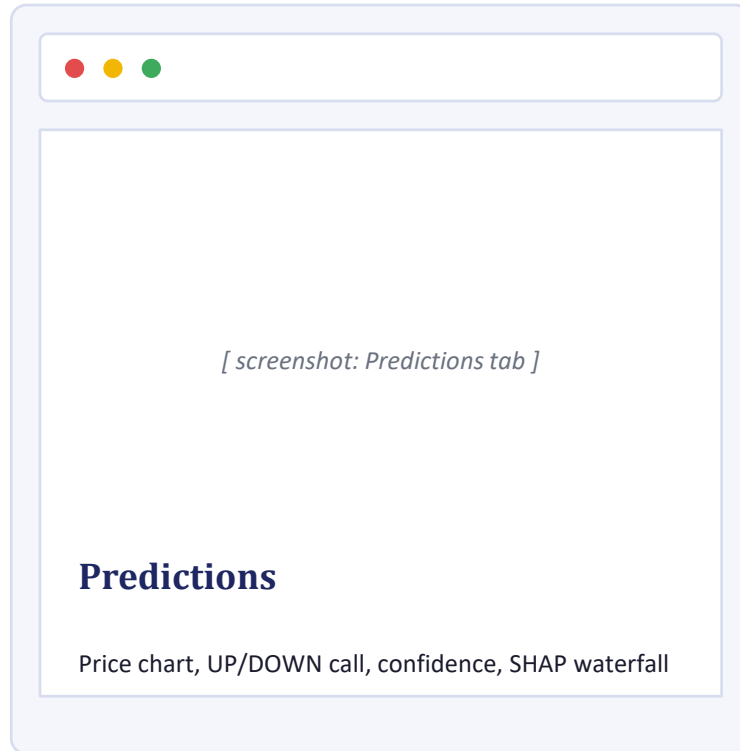
Example exchange

Q: “Which stock has the highest accuracy?”

A: “Based on the model results, TCS has the highest accuracy at 59%, followed by Sensex at 57%. This is retrieved directly from the stored evaluation metrics, not generated from memory.”

Streamlit App

Three tabs, one deployed URL



Insert actual screenshots from your deployed app before presenting.



Key Takeaways



Full pipeline, one codebase

Data acquisition → cleaning → features → modeling → explainability → RAG → deployment



Decisions, not just predictions

SHAP + cost-based thresholds turn raw probabilities into justified, business-aware calls



Conversational access

RAG lets any analyst query the system in plain English — no notebook required



Production-ready

Modular Python package + live Streamlit app with a public URL